Full Algebraic Derivation of 2D Radiation Diffusion Implicit Stencil

Contents

1 Continuous Equations in Cylindrical Coordinates

We start with the coupled radiation and material energy equations in axisymmetric cylindrical coordinates (r, z, t).

1.1 Radiation Energy Equation

The radiation energy density $E(r, z, t)$ evolves according to:

$$\frac{\partial E}{\partial t} = \frac{c}{3\sigma}\left(\frac{\partial^2 E}{\partial r^2} + \frac{1}{r}\frac{\partial E}{\partial r} + \frac{\partial^2 E}{\partial z^2}\right) - \chi c\sigma(E - U), \tag{1}$$

where

- c is the speed of light,
- $\sigma(T)$ is the temperature-dependent opacity (Rosseland mean), which varies in space via temperature,
- $U(r, z, t)$ is the material energy density (to be coupled below),
- the term $\frac{1}{r}\frac{\partial E}{\partial r}$ arises from cylindrical geometry.

Important: The form in Eq. (1) is written for notational simplicity. Since $\sigma(T(r, z))$ is *space-dependent*, the operator $\frac{c}{3\sigma}\nabla^2$ is not linear in the usual sense. In discretization, we must use the *conservative form* with diffusion coefficient $D(r, z) = \frac{c}{3\sigma(T(r,z))}$ varying at each location, giving:

$$\frac{\partial E}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rD\frac{\partial E}{\partial r}\right) + \frac{\partial}{\partial z}\left(D\frac{\partial E}{\partial z}\right) - \chi c\sigma(E - U). \tag{2}$$

This is the form we use throughout the derivation below.

1.2 Material Energy Equation

The material energy density $U(r, z, t)$ evolves according to:

$$\frac{1}{\beta}\frac{\partial U}{\partial t} = \chi c\sigma(E - U), \tag{3}$$

where $\beta = \frac{\partial U_R}{\partial U_m}$ is the ratio of radiation to material heat capacity (dimensionless coupling strength).

1.3 Decomposition of the Diffusion Operator

Define the space-dependent diffusion coefficient:

$$D(r, z, t) \equiv \frac{c}{3\sigma(T(r, z, t))}. \tag{4}$$

The diffusion operator in conservative form (Eq. (2)) is:

$$\mathcal{L}[E] \equiv \frac{1}{r}\frac{\partial}{\partial r}\left(rD\frac{\partial E}{\partial r}\right) + \frac{\partial}{\partial z}\left(D\frac{\partial E}{\partial z}\right) = \mathcal{L}_r[E] + \mathcal{L}_z[E], \tag{5}$$

where the radial part is:

$$\mathcal{L}_r[E] \equiv \frac{1}{r}\frac{\partial}{\partial r}\left(rD\frac{\partial E}{\partial r}\right) \tag{6}$$

and the axial part is:

$$\mathcal{L}_z[E] \equiv \frac{\partial}{\partial z}\left(D\frac{\partial E}{\partial z}\right). \tag{7}$$

In the numerical implementation, we discretize using a finite-volume approach where $D(r,z)$ is evaluated at cell faces and treated as piecewise constant over each control volume. The diffusion coefficients $D^n_{i,j\pm1/2}$ are then lagged in time (evaluated at t^n from U^n).

The coupled equations in conservative form are:

$$\frac{\partial E}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rD\frac{\partial E}{\partial r}\right) + \frac{\partial}{\partial z}\left(D\frac{\partial E}{\partial z}\right) - \chi c\sigma(E-U), \tag{8}$$

$$\frac{1}{\beta}\frac{\partial U}{\partial t} = \chi c\sigma(E-U). \tag{9}$$

2 Time Discretization via Backward Euler

We discretize time using the backward (implicit) Euler scheme on a sequence of times

$$t^n = t^{n-1} + \Delta t,$$

where $\Delta t = \Delta t_n$ may vary with n.

2.1 Implicit Euler for Radiation Energy

Apply backward Euler to Eq. (8):

$$\frac{E^{n+1}-E^n}{\Delta t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rD^n\frac{\partial E^{n+1}}{\partial r}\right) + \frac{\partial}{\partial z}\left(D^n\frac{\partial E^{n+1}}{\partial z}\right) - \chi c\sigma^n(E^{n+1}-U^{n+1}), \tag{10}$$

where we use time-lagged space-dependent coefficients $D^n(r,z)$ and $\sigma^n(r,z)$ (computed from U^n) at the old time, while spatial derivatives of E are evaluated at the new time (implicit in space, explicit in the coefficients).

Rearrange:

$$\frac{E^{n+1}}{\Delta t} - \frac{1}{r}\frac{\partial}{\partial r}\left(rD^n\frac{\partial E^{n+1}}{\partial r}\right) - \frac{\partial}{\partial z}\left(D^n\frac{\partial E^{n+1}}{\partial z}\right) + \chi c\sigma^n E^{n+1} - \chi c\sigma^n U^{n+1} = \frac{E^n}{\Delta t}. \tag{11}$$

2.2 Implicit Euler for Material Energy

Note on LTE coupling: The coupling factor χ (dimensionless) modulates the interaction strength between radiation and material in Local Thermodynamic Equilibrium (LTE). The factor appears in $A^n = \beta^n \Delta t \cdot \chi \cdot c\sigma^n$, with typical values $\chi = 1$ for full coupling or $\chi < 1$ for reduced coupling.

Apply backward Euler to Eq. (9), using time-lagged coefficients β^n and σ^n (computed from U^n):

$$\frac{1}{\beta^n}\frac{U^{n+1} - U^n}{\Delta t} = \chi c\sigma^n (E^{n+1} - U^{n+1}). \tag{12}$$

Rearrange:

$$\frac{1}{\beta^n} U^{n+1} + \Delta t \cdot \chi \cdot c\sigma^n U^{n+1} = \frac{1}{\beta^n} U^n + \Delta t \cdot \chi \cdot c\sigma^n E^{n+1}. \tag{13}$$

Factor:

$$\left(\frac{1}{\beta^n} + \Delta t \cdot \chi \cdot c\sigma^n\right) U^{n+1} = \frac{1}{\beta^n} U^n + \Delta t \cdot \chi \cdot c\sigma^n E^{n+1}. \tag{14}$$

Define:

$$A^n \equiv \beta^n \Delta t \cdot \chi \cdot c\sigma^n. \tag{15}$$

Then:

$$(1 + A^n)\, U^{n+1} = U^n + A^n E^{n+1}. \tag{16}$$

Solve for U^{n+1} in terms of E^{n+1}:

$$U^{n+1} = \frac{U^n + A^n E^{n+1}}{1 + A^n} \tag{17}$$

2.3 Substituting the Material Update into the Radiation Equation

Substitute Eq. (17) into the radiation equation:

$$\frac{E^{n+1}}{\Delta t} - \frac{1}{r}\frac{\partial}{\partial r}\left(r D^n \frac{\partial E^{n+1}}{\partial r}\right) - \frac{\partial}{\partial z}\left(D^n \frac{\partial E^{n+1}}{\partial z}\right) + \chi c\sigma^n E^{n+1} - \chi c\sigma^n \frac{U^n + A^n E^{n+1}}{1 + A^n} = \frac{E^n}{\Delta t}. \tag{18}$$

Expand the coupling term:

$$\chi c\sigma^n \frac{U^n + A^n E^{n+1}}{1 + A^n} = \frac{\chi c\sigma^n U^n}{1 + A^n} + \frac{\chi c\sigma^n A^n E^{n+1}}{1 + A^n}. \tag{19}$$

Substitute back:

$$\frac{E^{n+1}}{\Delta t} - \frac{1}{r}\frac{\partial}{\partial r}\left(r D^n \frac{\partial E^{n+1}}{\partial r}\right) - \frac{\partial}{\partial z}\left(D^n \frac{\partial E^{n+1}}{\partial z}\right) + \chi c\sigma^n E^{n+1} - \frac{\chi c\sigma^n A^n E^{n+1}}{1 + A^n} = \frac{E^n}{\Delta t} + \frac{\chi c\sigma^n U^n}{1 + A^n}. \tag{20}$$

Combine the E^{n+1} coefficients on the left:

$$\frac{1}{\Delta t} + \chi c\sigma^n - \frac{\chi c\sigma^n A^n}{1 + A^n}. \tag{21}$$

Simplify the coupling diagonal:

$$\chi c\sigma^n - \frac{\chi c\sigma^n A^n}{1+A^n} = \chi c\sigma^n \left(1 - \frac{A^n}{1+A^n}\right) \tag{22}$$

$$= \chi c\sigma^n \left(\frac{1+A^n-A^n}{1+A^n}\right) \tag{23}$$

$$= \frac{\chi c\sigma^n}{1+A^n}. \tag{24}$$

So the equation becomes:

$$\left(\frac{1}{\Delta t} + \frac{\chi c\sigma^n}{1+A^n}\right) E^{n+1} - -\frac{1}{r}\frac{\partial}{\partial r}\left(rD^n \frac{\partial E^{n+1}}{\partial r}\right) - \frac{\partial}{\partial z}\left(D^n \frac{\partial E^{n+1}}{\partial z}\right) = \frac{E^n}{\Delta t} + \frac{\chi c\sigma^n}{1+A^n}U^n. \tag{25}$$

Define the effective coupling coefficient:

$$C^n \equiv \frac{\chi c\sigma^n}{1+A^n}. \tag{26}$$

Then Eq. (25) reads:

$$\left(\frac{1}{\Delta t} + C^n\right) E^{n+1} - \frac{1}{r}\frac{\partial}{\partial r}\left(rD^n \frac{\partial E^{n+1}}{\partial r}\right) - \frac{\partial}{\partial z}\left(D^n \frac{\partial E^{n+1}}{\partial z}\right) = \frac{E^n}{\Delta t} + C^n U^n. \tag{27}$$

3 Spatial Discretization

3.1 Grid Definition

We discretize the domain into a 2D grid of cell-centers:

$$z_i = (i-1)\,\Delta z, \quad i = 0, 1, \ldots, N_z - 1, \quad \Delta z = \frac{L_z}{N_z - 1}, \tag{28}$$

$$r_j = \text{general radial nodes}, \quad j = 0, 1, \ldots, N_r - 1. \tag{29}$$

In general, r_j may be non-uniform (e.g., uniform in the foam region and geometrically stretched in the gold layer). For simplicity in the derivation, we first assume uniform Δr, then note the modifications for non-uniform grids.

Stored fields are cell-centered:

$$E^n_{i,j} \approx E(r_j, z_i, t^n), \quad U^n_{i,j} \approx U(r_j, z_i, t^n). \tag{30}$$

3.2 Face Values and Averaging

Spatial derivatives are computed at cell faces. The faces are located at:

$$r_{j-\frac{1}{2}} = \frac{r_{j-1} + r_j}{2}, \quad \Delta r_j = r_j - r_{j-1}, \tag{31}$$

$$z_{i-\frac{1}{2}} = \frac{z_{i-1} + z_i}{2}, \quad \Delta z = z_i - z_{i-1}. \tag{32}$$

For space-dependent coefficients like $D(T(r,z))$, we evaluate T at cell centers to get $D_{i,j}$, then average D to each face. Common averaging methods are:

$$\text{Arithmetic:} \quad D_{i,j+\frac{1}{2}} \approx \frac{D_{i,j}+D_{i,j+1}}{2}, \tag{33}$$

$$\text{Harmonic:} \quad D_{i,j+\frac{1}{2}} \approx \frac{2D_{i,j}D_{i,j+1}}{D_{i,j}+D_{i,j+1}}, \tag{34}$$

$$\text{Geometric:} \quad D_{i,j+\frac{1}{2}} \approx \sqrt{D_{i,j}D_{i,j+1}}. \tag{35}$$

The harmonic mean is often preferred for diffusion coefficients to ensure monotonicity, but the code supports all three. Similarly for z-faces. These face-averaged coefficients $D^n_{i,j\pm 1/2}$ are then frozen at the old time t^n during the implicit solve.

4 Radial Diffusion Operator

4.1 Conservative Form

The radial part of the diffusion operator with space-dependent $D(r,z)$ is:

$$\mathcal{L}_r[E] = \frac{1}{r}\frac{d}{dr}\left(rD\frac{\partial E}{\partial r}\right). \tag{36}$$

This is the correct conservative form when D varies with position. We discretize by defining radial fluxes at cell faces:

$$F^r_{i,j+\frac{1}{2}} \equiv r_{j+\frac{1}{2}}\, D^n_{i,j+\frac{1}{2}}\, \frac{E^{n+1}_{i,j+1}-E^{n+1}_{i,j}}{\Delta r_{j+\frac{1}{2}}}, \quad \Delta r_{j+\frac{1}{2}} = r_{j+1}-r_j. \tag{37}$$

The divergence over a control volume at (i,j) is then:

$$\left[\frac{1}{r}\frac{d}{dr}\left(rD\frac{\partial E}{\partial r}\right)\right]_{i,j} \approx \frac{1}{r_j\,\Delta r_j}\left(F^r_{i,j+\frac{1}{2}}-F^r_{i,j-\frac{1}{2}}\right), \tag{38}$$

where Δr_j is the width of the control volume around node j (typically the average of $\Delta r_{j-\frac{1}{2}}$ and $\Delta r_{j+\frac{1}{2}}$).

4.2 Discrete Form for Interior Nodes

For interior nodes $1 \le j \le N_r - 2$, expanding the space-dependent operator:

$$\left[\frac{1}{r}\frac{d}{dr}\left(rD\frac{\partial E}{\partial r}\right)\right]_{i,j} \approx \frac{1}{r_j\,\Delta r_j}\left[r_{j+\frac{1}{2}}\, D^n_{i,j+\frac{1}{2}}\, \frac{E^{n+1}_{i,j+1}-E^{n+1}_{i,j}}{\Delta r_{j+\frac{1}{2}}} - r_{j-\frac{1}{2}}\, D^n_{i,j-\frac{1}{2}}\, \frac{E^{n+1}_{i,j}-E^{n+1}_{i,j-1}}{\Delta r_{j-\frac{1}{2}}}\right]. \tag{39}$$

Here, $D^n_{i,j+\frac{1}{2}}$ is the face-averaged diffusion coefficient at time t^n. Rearrange as a linear combination:

$$\approx \frac{r_{j+\frac{1}{2}}\, D^n_{i,j+\frac{1}{2}}}{r_j\,\Delta r_j\,\Delta r_{j+\frac{1}{2}}}E^{n+1}_{i,j+1} - \frac{r_{j+\frac{1}{2}}\, D^n_{i,j+\frac{1}{2}}}{r_j\,\Delta r_j\,\Delta r_{j+\frac{1}{2}}}E^{n+1}_{i,j} - \frac{r_{j-\frac{1}{2}}\, D^n_{i,j-\frac{1}{2}}}{r_j\,\Delta r_j\,\Delta r_{j-\frac{1}{2}}}E^{n+1}_{i,j} + \frac{r_{j-\frac{1}{2}}\, D^n_{i,j-\frac{1}{2}}}{r_j\,\Delta r_j\,\Delta r_{j-\frac{1}{2}}}E^{n+1}_{i,j-1}. \tag{40}$$

Define the effective cylindrical weights (with space-dependent coefficients):

$$\lambda^{r,n}_{i,j+\frac{1}{2}} \equiv \frac{r_{j+\frac{1}{2}}\, D^n_{i,j+\frac{1}{2}}}{r_j\, \Delta r_j\, \Delta r_{j+\frac{1}{2}}}, \tag{41}$$

$$\lambda^{r,n}_{i,j-\frac{1}{2}} \equiv \frac{r_{j-\frac{1}{2}}\, D^n_{i,j-\frac{1}{2}}}{r_j\, \Delta r_j\, \Delta r_{j-\frac{1}{2}}}. \tag{42}$$

Then the discrete radial contribution to the implicit step is:

$$\left[\frac{1}{r}\frac{d}{dr}\left(rD^n\frac{\partial E}{\partial r}\right)\right]^{n+1}_{i,j} \approx \lambda^{r,n}_{i,j+\frac{1}{2}} E^{n+1}_{i,j+1} - \left(\lambda^{r,n}_{i,j+\frac{1}{2}} + \lambda^{r,n}_{i,j-\frac{1}{2}}\right) E^{n+1}_{i,j} + \lambda^{r,n}_{i,j-\frac{1}{2}} E^{n+1}_{i,j-1}. \tag{43}$$

5 Axial Diffusion Operator

5.1 Continuous Form

The axial part with space-dependent $D(r,z)$:

$$\mathcal{L}_z[E] = \frac{\partial}{\partial z}\left(D\frac{\partial E}{\partial z}\right). \tag{44}$$

5.2 Discrete Form for Interior Nodes

For interior axial nodes $i = 1, \ldots, N_z - 2$, the space-dependent operator is approximated as:

$$\left[\frac{\partial}{\partial z}\left(D\frac{\partial E}{\partial z}\right)\right]_{i,j} \approx \frac{1}{(\Delta z)^2}\left[D^n_{i+1,j}\, E^{n+1}_{i+1,j} - (D^n_{i+1,j} + D^n_{i,j})\, E^{n+1}_{i,j} + D^n_{i,j}\, E^{n+1}_{i-1,j}\right]. \tag{45}$$

However, for simplicity and to maintain stability, a common approximation averages the diffusion coefficient to the cell center:

$$D^n_{i,j} \approx \frac{D^n_{i+1,j} + D^n_{i,j} + D^n_{i-1,j}}{3}, \tag{46}$$

which then gives the simpler form:

$$\left[\frac{\partial}{\partial z}\left(D\frac{\partial E}{\partial z}\right)\right]_{i,j} \approx D^n_{i,j}\, \frac{E^{n+1}_{i+1,j} - 2E^{n+1}_{i,j} + E^{n+1}_{i-1,j}}{(\Delta z)^2}. \tag{47}$$

With implicit time stepping:

$$\left[\frac{\partial}{\partial z}\left(D^n\frac{\partial E}{\partial z}\right)\right]^{n+1}_{i,j} \approx \frac{D^n_{i,j}}{(\Delta z)^2}\left[E^{n+1}_{i+1,j} - 2E^{n+1}_{i,j} + E^{n+1}_{i-1,j}\right]. \tag{48}$$

Define the axial weight:

$$\lambda^{z,n}_{i,j} \equiv \frac{D^n_{i,j}}{(\Delta z)^2}. \tag{49}$$

Then:

$$\left[D\frac{\partial^2 E}{\partial z^2}\right]^{n+1}_{i,j} \approx \lambda^{z,n}_{i,j} E^{n+1}_{i+1,j} - 2\lambda^{z,n}_{i,j} E^{n+1}_{i,j} + \lambda^{z,n}_{i,j} E^{n+1}_{i-1,j}. \tag{50}$$

6 Complete Interior 5-Point Stencil

6.1 Combining All Terms

Start from Eq. (27) and apply time discretization:

$$\left(\frac{1}{\Delta t} + C^n\right) E_{i,j}^{n+1} - \left[\mathcal{L}_r + \mathcal{L}_z\right] E_{i,j}^{n+1} = \frac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n. \tag{51}$$

Substitute the discrete forms from Eqs. (43) and (50):

$$\left(\frac{1}{\Delta t} + C^n\right) E_{i,j}^{n+1} - \left[\lambda_{i,j+\frac{1}{2}}^{r,n} E_{i,j+1}^{n+1} - (\lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n}) E_{i,j}^{n+1} + \lambda_{i,j-\frac{1}{2}}^{r,n} E_{i,j-1}^{n+1}\right] \tag{52}$$

$$- \left[\lambda_{i,j}^{z,n} E_{i+1,j}^{n+1} - 2\lambda_{i,j}^{z,n} E_{i,j}^{n+1} + \lambda_{i,j}^{z,n} E_{i-1,j}^{n+1}\right] = \frac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n. \tag{53}$$

Collect all terms on the left:

$$\left(\frac{1}{\Delta t} + C^n + \lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n} + 2\lambda_{i,j}^{z,n}\right) E_{i,j}^{n+1} \tag{54}$$

$$- \lambda_{i,j+\frac{1}{2}}^{r,n} E_{i,j+1}^{n+1} - \lambda_{i,j-\frac{1}{2}}^{r,n} E_{i,j-1}^{n+1} - \lambda_{i,j}^{z,n} E_{i+1,j}^{n+1} - \lambda_{i,j}^{z,n} E_{i-1,j}^{n+1} \tag{55}$$

$$= \frac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n. \tag{56}$$

Rearrange into standard form:

$$\boxed{\begin{array}{ll} \text{diagonal coeff:} & \dfrac{1}{\Delta t} + C^n + \lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n} + 2\lambda_{i,j}^{z,n}, \\ \text{r-neighbors:} & -\lambda_{i,j\pm\frac{1}{2}}^{r,n}, \\ \text{z-neighbors:} & -\lambda_{i,j}^{z,n}, \\ \text{RHS:} & \dfrac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n. \end{array}} \tag{57}$$

This is the full interior 5-point stencil (or 4-point in 1D) used in the implicit solver.

7 Boundary Conditions

7.1 Axial Boundary Conditions

7.1.1 Dirichlet Boundary at $z = 0$ (Drive)

At the irradiated surface, we impose:

$$E_{0,j}^{n+1} = E_{\text{drive}}(t^{n+1}, j), \tag{58}$$

where $E_{\text{drive}}(t)$ is computed from the imposed drive temperature $T_D(t)$ via $E_{\text{drive}} = aT_D^4$ with $a = 4\sigma_{\text{SB}}/c$.

This node is eliminated from the linear system; instead, its value is imposed after the solve.

7.1.2 Marshak Boundary at $z = 0$

Alternatively, the Marshak boundary condition relates the boundary value to the outward flux:

$$aT_D^4(t) = E(0, j, t) - \frac{2D(0, j, t)}{c} \left.\frac{\partial E}{\partial z}\right|_{z=0}. \tag{59}$$

Using backward differencing:

$$\left.\frac{\partial E}{\partial z}\right|_{z=0} \approx \frac{E_{1,j}^{n+1} - E_{0,j}^{n+1}}{\Delta z}. \tag{60}$$

Substituting:

$$aT_D^4(t^{n+1}) = E_{0,j}^{n+1} - \frac{2D_{0,j}^n}{c} \frac{E_{1,j}^{n+1} - E_{0,j}^{n+1}}{\Delta z}. \tag{61}$$

Multiply by Δz and rearrange:

$$aT_D^4(t^{n+1})\Delta z = E_{0,j}^{n+1}\Delta z - \frac{2D_{0,j}^n}{c} (E_{1,j}^{n+1} - E_{0,j}^{n+1}). \tag{62}$$

$$aT_D^4(t^{n+1})\Delta z = E_{0,j}^{n+1} \left(\Delta z + \frac{2D_{0,j}^n}{c}\right) - \frac{2D_{0,j}^n}{c} E_{1,j}^{n+1}. \tag{63}$$

Define $\alpha_j = \frac{2D_{0,j}^n}{c\,\Delta z}$. Then:

$$aT_D^4(t^{n+1}) = (1 + \alpha_j)\, E_{0,j}^{n+1} - \alpha_j\, E_{1,j}^{n+1}. \tag{64}$$

This is the discrete Marshak row at $z = 0$, replacing the interior 5-point stencil for that location.

7.1.3 Dirichlet Boundary at $z = L_z$ (Bath)

At the far end:

$$E_{N_z-1,j}^{n+1} = E_{\text{bath}} = a(300\text{ K})^4. \tag{65}$$

7.2 Radial Boundary Conditions

7.2.1 Axis Symmetry at $r = 0$

By cylindrical symmetry:

$$\left.\frac{\partial E}{\partial r}\right|_{r=0} = 0. \tag{66}$$

This is enforced by setting $E_{i,0} = E_{i,1}$, which removes the $1/r$ singularity.

7.2.2 Dirichlet Boundary at $r = R_{\text{tot}}$ (Fixed Bath)

Impose:

$$E_{i,N_r-1}^{n+1} = E_{\text{bath}}. \tag{67}$$

7.2.3 Neumann Boundary at $r = R_{\text{tot}}$ (No Flux)

Impose:

$$\left.\frac{\partial E}{\partial r}\right|_{r=R_{\text{tot}}} = 0, \tag{68}$$

which eliminates the outer-face flux contribution in the stencil.

7.2.4 Marshak Boundary at $r = R_{\text{tot}}$ (Radiating Wall)

The Marshak condition at the outer surface is:

$$aT_{\text{wall}}^4(t) = E(N_r - 1, i, t) - \frac{2D(N_r - 1, i, t)}{c} \left.\frac{\partial E}{\partial r}\right|_{r=R_{\text{tot}}}. \tag{69}$$

Using backward differencing:

$$\left.\frac{\partial E}{\partial r}\right|_{r=R_{\text{tot}}} \approx \frac{E_{i,N_r-1}^{n+1} - E_{i,N_r-2}^{n+1}}{\Delta r_{\text{outer}}}. \tag{70}$$

Substituting and rearranging similarly to Eq. (64):

$$\boxed{aT_{\text{wall}}^4(t^{n+1}) = (1 + \alpha_i)\, E_{i,N_r-1}^{n+1} - \alpha_i\, E_{i,N_r-2}^{n+1},} \tag{71}$$

where $\alpha_i = \frac{2D_{i,N_r-1}^n}{c\,\Delta r_{\text{outer}}}$.

This row is included in the linear system to replace the interior 5-point stencil at the outer radial boundary.

8 Linear System Assembly

8.1 Unknown Vector Organization

The implicit solve solves for E^{n+1} on an interior subset of the grid. Typically:

- If Marshak boundary at $z = 0$: unknowns are $E_{i,j}^{n+1}$ for $i = 0, \ldots, N_z - 2$ and $j = 0, \ldots, N_r - 1$.
- If Dirichlet at $z = 0$: unknowns are $E_{i,j}^{n+1}$ for $i = 1, \ldots, N_z - 2$ and $j = 0, \ldots, N_r - 1$.
- Boundary nodes at $z = L_z$ and possibly $r = R$ are either imposed after the solve or handled specially.

The total number of unknowns is:

$$n_{\text{unknown}} = n_z \times n_r, \tag{72}$$

where n_z and n_r are the counts of unknown rows in each direction.

8.2 Row-Wise Assembly

For each unknown (i,j), the corresponding matrix row contains:

- Diagonal entry: coefficient of $E_{i,j}^{n+1}$,
- Off-diagonal entries: coefficients of neighboring $E_{i\pm1,j}^{n+1}$ and $E_{i,j\pm1}^{n+1}$,
- RHS: constant vector entries.

The sparsity pattern is structured: each row has at most 5 nonzeros (2D grid), or fewer at boundaries.

8.3 CSR Storage

To store the matrix efficiently in Compressed Sparse Row (CSR) format:

$$\text{indptr}: \text{ row start pointers (length } n_{\text{unknown}} + 1), \tag{73}$$

$$\text{indices}: \text{ column indices of stored values,} \tag{74}$$

$$\text{data}: \text{ numeric coefficients.} \tag{75}$$

For each row, we store only the nonzero entries in sorted column order, reducing memory from $O(n_{\text{unknown}}^2)$ dense to $O(n_{\text{unknown}})$ sparse.

9 Material Energy Update

After solving for E^{n+1}, update U^{n+1} using Eq. (17):

$$U_{i,j}^{n+1} = \frac{U_{i,j}^n + A_{i,j}^n E_{i,j}^{n+1}}{1 + A_{i,j}^n}, \tag{76}$$

where $A_{i,j}^n = \beta_{i,j}^n \Delta t \cdot \chi \cdot c\sigma_{i,j}^n$.

This is a local (cell-by-cell) update, not requiring a linear solve.

10 Self-Similar Model Integration

In the self-similar model, the coupling parameter $\beta(T)$ (hence β^n at the old time) and the heat capacity are related to material parameters via:

$$\beta(T) = \mathrm{Cv}_R(T)/\mathrm{Cv}_m(T), \tag{77}$$

which is the ratio of radiation to material heat capacity (dimensionless). This is temperature-dependent and must be evaluated at each location from the old material energy U^n to get $\beta_{i,j}^n$.

The opacity and heat capacity are functions of temperature:

$$\sigma(T) = \frac{1}{gT^{\alpha}\rho^{-\lambda-1}}, \tag{78}$$

$$\mathrm{Cv}_m(T) = f\beta_{\mathrm{exp}}T^{\beta_{\mathrm{exp}}-1}\rho^{-\mu+1}, \tag{79}$$

$$\mathrm{Cv}_R(T) = 4aT^3. \tag{80}$$

At each timestep, $T^n = (U^n/a)^{1/4}$ is computed from the old material energy, and all temperature-dependent quantities (D^n, σ^n, C^n) are evaluated there.

A Quick Reference: Final Stencil Summary

A.1 Interior 5-Point Stencil

For interior nodes (i, j) with $1 \le i \le N_z - 2$ and $1 \le j \le N_r - 2$ (or adjusted ranges for Marshak):

A.1.1 Matrix Coefficients

$$a_{\mathrm{self}} = \frac{1}{\Delta t} + C^n + \lambda^{r,n}_{i,j+\frac{1}{2}} + \lambda^{r,n}_{i,j-\frac{1}{2}} + 2\lambda^{z,n}_{i,j}, \tag{81}$$

$$a_{i+1,j} = -\lambda^{z,n}_{i,j}, \tag{82}$$

$$a_{i-1,j} = -\lambda^{z,n}_{i,j}, \tag{83}$$

$$a_{i,j+1} = -\lambda^{r,n}_{i,j+\frac{1}{2}}, \tag{84}$$

$$a_{i,j-1} = -\lambda^{r,n}_{i,j-\frac{1}{2}}. \tag{85}$$

A.1.2 RHS

$$b_{i,j} = \frac{E^n_{i,j}}{\Delta t} + C^n_{i,j}U^n_{i,j}, \tag{86}$$

where $C^n_{i,j} = \frac{c\sigma^n_{i,j}}{1+A^n_{i,j}}$ and $A^n_{i,j} = \beta^n_{i,j}\Delta t \cdot \chi \cdot c\sigma^n_{i,j}$.

A.2 Marshak Rows at $z = 0$

$$a_{\mathrm{self}} = 1 + \alpha_j, \quad \alpha_j = \frac{2D^n_{0,j}}{c\,\Delta z}, \tag{87}$$

$$a_{1,j} = -\alpha_j, \tag{88}$$

$$b_{0,j} = aT_D^4(t^{n+1}). \tag{89}$$

A.3 Marshak Rows at $r = R_{\text{tot}}$

$$a_{\text{self}} = 1 + \alpha_i, \quad \alpha_i = \frac{2D^n_{i,N_r-1}}{c\,\Delta r_{\text{outer}}}, \tag{90}$$

$$a_{i,N_r-2} = -\alpha_i, \tag{91}$$

$$b_{i,N_r-1} = aT^4_{\text{wall}}(t^{n+1}). \tag{92}$$